

Engineering Mathematics Study Note: Directional Derivative Between Two Points

1. IDENTIFY

The concept being executed is finding the **Directional Derivative** of a scalar field ϕ in the specific direction connecting an initial point A to a terminal point B in 3D space.

2. THE FORMULA

The Direction Vector $\mathbf{a}$ from point $A(x_1, y_1, z_1)$ to point $B(x_2, y_2, z_2)$ is found by coordinate subtraction:

$$\mathbf{a} = \vec{AB} = (x_2 - x_1)\mathbf{i} + (y_2 - y_1)\mathbf{j} + (z_2 - z_1)\mathbf{k}$$

The Directional Derivative (D.D) of ϕ along vector $\mathbf{a}$ is:

$$\text{D.D} = \frac{\nabla\phi \cdot \mathbf{a}}{|\mathbf{a}|}$$

Where the Gradient vector evaluated at the starting point A is:

$$\nabla\phi = \mathbf{i}\frac{\partial\phi}{\partial x} + \mathbf{j}\frac{\partial\phi}{\partial y} + \mathbf{k}\frac{\partial\phi}{\partial z}$$

3. THE ALGORITHM

Step 1: Compute the partial derivatives

Find the derivative of ϕ with respect to x , y , and z independently, treating all other variables as numbers.

Step 2: Construct and evaluate the Gradient vector at point A

Assemble $\nabla\phi$ and substitute the coordinates of the *starting point* $A(1, -2, 1)$ into it.

Step 3: Find the displacement direction vector $\mathbf{a}$

Subtract the coordinates of starting point A from destination point B :

$$\mathbf{a} = \vec{AB} = \text{Vector}(B - A)$$

Step 4: Calculate the magnitude of the displacement vector

Find the geometric length of the direction vector $\mathbf{a}$.

$$|\mathbf{a}| = \sqrt{(a_x)^2 + (a_y)^2 + (a_z)^2}$$

Step 5: Perform the Dot Product and scale

Multiply the matching directional components of the evaluated gradient vector and vector $\mathbf{a}$ together, add them up, and divide by the magnitude found in Step 4.

4. THE MECHANICS

Vector from Point A to Point B ($\vec{AB}$)

To find the vector path between two points, always subtract the *start* from the *finish* ($B - A$).

- If you start at $x = 1$ and finish at $x = 2$, your displacement is $2 - 1 = 1$.
- If you start at $y = -2$ and finish at $y = 6$, your displacement is $6 - (-2) = 6 + 2 = 8$.

Cubing Negative Numbers

When evaluating the gradient component for an odd power like y^3 where $y = -2$:

- $(-2)^3 = (-2) \times (-2) \times (-2) = -8$
 - Multiplying by the constant multiplier 4: $4 \times (-8) = -32$
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5. STEP-BY-STEP WORKED SOLUTION

Execution of Step 1: Compute Partial Derivatives

$$\frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x}(x^4 + y^4 + z^4) = 4x^3$$

$$\frac{\partial \phi}{\partial y} = \frac{\partial}{\partial y}(x^4 + y^4 + z^4) = 4y^3$$

$$\frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z}(x^4 + y^4 + z^4) = 4z^3$$

Execution of Step 2: Evaluate Gradient at $A(1, -2, 1)$

$$\nabla \phi = 4x^3 \mathbf{i} + 4y^3 \mathbf{j} + 4z^3 \mathbf{k}$$

$$\nabla \phi_{(1,-2,1)} = 4(1)^3 \mathbf{i} + 4(-2)^3 \mathbf{j} + 4(1)^3 \mathbf{k}$$

$$\nabla \phi_{(1,-2,1)} = 4(1) \mathbf{i} + 4(-8) \mathbf{j} + 4(1) \mathbf{k}$$

$$\nabla \phi_{(1,-2,1)} = 4\mathbf{i} - 32\mathbf{j} + 4\mathbf{k}$$

Execution of Step 3: Compute Direction Vector $\mathbf{a} = \vec{AB}$

Given $A(1, -2, 1)$ and $B(2, 6, -1)$:

$$\mathbf{a} = (2 - 1)\mathbf{i} + (6 - (-2))\mathbf{j} + (-1 - 1)\mathbf{k}$$

$$\mathbf{a} = 1\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}$$

Execution of Step 4: Calculate Magnitude $|\mathbf{a}|$

$$|\mathbf{a}| = \sqrt{1^2 + 8^2 + (-2)^2}$$

$$|\mathbf{a}| = \sqrt{1 + 64 + 4}$$

$$|\mathbf{a}| = \sqrt{69}$$

Execution of Step 5: Dot Product and Final Calculation

$$D.D = \frac{(4\mathbf{i} - 32\mathbf{j} + 4\mathbf{k}) \cdot (1\mathbf{i} + 8\mathbf{j} - 2\mathbf{k})}{\sqrt{69}}$$

$$D.D = \frac{(4 \times 1) + (-32 \times 8) + (4 \times -2)}{\sqrt{69}}$$

$$D.D = \frac{4 - 256 - 8}{\sqrt{69}}$$

$$D.D = \frac{-260}{\sqrt{69}}$$